# Heat Transfer

Final Lab Report
Unit Operations Lab 1
February 2, 2026

Group 1
*Ify Bidokwu*
*Cesearo Dilosa*
*Javier Silva*
*Joshua Laforest*

## 1. Executive Summary

An experimental study was conducted to investigate heat transfer behavior in a jacketed stirred tank and to explore the influence of impeller and baffle configuration on system performance. The objective of the experiment was to evaluate how different mixing conditions can affect the heat transfer process in industrial equipment.

The apparatus consisted of a jacketed stirred tank equipped with an impeller, removable baffles, thermometers, external heat exchanger, internal cooling coil, flow meters, a steam line, steam trap, pump, and a drain. Experiments were performed at different impeller speeds ranging from 300-900 RPM, the trials were conducted with and without baffles installed. At steady state, the flow rates of the coiling coil water and recirculation flow were collected. Additionally, the steam condensate was collected from the steam condenser and measured to determine the steam flowrate. Temperatures were collected from the readings from the multiple temperature gauges from the system.

From the data collected, higher impeller speeds corresponded to increased heat transfer activity. Trends in the calculated heat transfer coefficients demonstrate higher rotation speed resulting in larger heat transfer coefficients. In most cases, baffled runs exhibited higher heat transfer performance compared to unbaffled runs.

Evidenced by the results, there is a strong dependence of heat transfer based on vessel configuration and agitation. Flow turbulence can enhance energy exchange between fluid and surface. The addition of baffles causes the flow to flow perpendicularly omitting the existence of dead zones, increasing efficiency.

## 2. Introduction

The objective of the experiment was to analyze heat transfer in a stirred tank by determining the heat transfer coefficients across a vessel wall, both with and without baffles, at different impeller speeds. The apparatus consists of several components, including a tank, stirrer, heat exchangers, etc.

The significance of the experiment lies in understanding how variations in mixing conditions, impeller speed and baffle configuration can influence heat transfer. These parameters form the basis of thermal design in industries such as biochemical engineering, pharmaceutical, and food manufacturing. In the food industry, jacket stirred tanks are used to eliminate bacteria in fluid-based foods while minimizing quality loss (*Fryer 2004*). Similarly, the pharmaceutical industry employs stirred tank reactors for vaccine and drug production to ensure purity during chemical synthesis as they regulate temperature and mixing speed. (*Antognoli 2025*).

## 3. Theory

The concept of heat transfer across a solid wall using two fluids involves a series of thermal resistances. At a steady state, the heat transferred is continuous across the interface. For this specific stirred tank, the process involves a heat transfer from the condensed steam in the steam jacket, then conduction through the metal wall and finally, convection from the wall to the vessel fluid (*Heat Transfer Lab Manual Page 1 & 2*). For this experiment, the team assumed no scale formation and negligible metal wall resistance. The main resistances are the hot and cold side convection films.

At steady state, the total heat transfer rate through the interface is written using the overall heat transfer coefficient (*U*) and the temperature difference between both steam and tank water (*Heat Transfer Lab Manual Page 3*).

The overall heat transfer coefficient is related to the individual film coefficients for the tank and steam side surfaces (*Heat Transfer Lab Manual Page 3*).

To find $Q_{13}$ experimentally, a steady-state heat balance is added where the heat added equals the heat removed. Using the 2nd method, the heat is removed through the external heat exchanger, the internal cooling coils, and heat loss to the surroundings (*Heat Transfer Lab Manual Page 4 & 5*).

The heat removed by the external heat exchanger is gotten from the recirculation flow rate and temperature drop (*Heat Transfer Lab Manual Page 4 & 5*).

The heat removed by the internal cooling coil is gotten from the flow rate of the coil and temperature rise

**Empirical Correlations**

The tank side heat transfer coefficient can be compared to the theoretical value predicted using a stirred-tank Nusselt number correlation (*Heat Transfer Lab Manual Page 2,5 & 9*).

The dimensionless groups are stated as:

## 4. Results

The main objective of this experiment, in terms of data, was to find some correlation between the independent variables, impeller rpm & recirculation flowrate, vs dependent variables, overall heat transfer coefficient ($U_i$) & the standard heat transfer coefficients ($h_i$, $h_o$). The effect of $Q_{13}$ on the chosen dependent variables was also examined. For this lab, 3 trials at various impeller rpms ranging from 300-900 were ran for the baffled and unbaffled configuration. The recirculation flowrate (as well as the cooling coil and steam condensate flowrates) was varied to keep the tank temperature between 65 ℃ and 85 ℃. This in turn caused $Q_{13}$ to also change throughout the experiments.

*Note: For the entire results section, $h_i$ was calculated using equation (9) from the lab handout because it gave results that were closer to literature values for similar lab set-ups [3]*

**Impeller RPM vs. $U_i$, $h_i$, and $h_o$:**

To begin this lab, data was gathered for the unbaffled tank first, then the baffled. For both configurations, we wanted to begin by determining the effect of the impeller speed on the overall heat transfer coefficient, $U_i$, and the standard heat transfer coefficients ($h_i$, $h_o$). Once data was gathered, plots were made to determine a relationship. For $U_i$, there was a noticeable non-linear downtrend as impeller speed increased from ~300 rpm to ~900 rpm. In contrast, $h_i$ and $h_o$ saw uptrends as impeller rpm increased; $h_o$'s values decreased non-linearly while $h_i$ saw a linear increase. Overall, there wasn't much distinction between baffled and unbaffled. When measuring $U_i$, $h_i$, or $h_o$ sometimes the values would be higher for the baffled tank and other times their values would be higher for the unbaffled tank.

The calculation of $U_i$ was accomplished using equation (3) from the lab handout [5]. To use this equation, the surface area of the tank needed to be calculated, and the temperature of the steam had to be calculated for each trial. To calculate the steam temperature, I used the value of $Q_{13}$, which was previously determined, along with the $C_p$ value of steam at ~100 ℃ as a good estimate [4]. To calculate $h_o$, equation (13) was used. To calculate $h_i$, equations (4) and (9) were used, respectively, to compare values. Graphs of these trends, along with tabled data, are presented in Appendix I, while sample calculations are presented in Appendix III at the end of this report. The units for $U_i$, $h_i$, and $h_o$ were $\frac{W}{m^2 * K}$. The unit for $Q_{13}$ was kW, while the unit for impeller revolutions was rpm.

**Recirculation Flowrate vs. $U_i$, $h_i$, and $h_o$:**

Next, the effect of the recirculation flowrate of the water leaving the external heat exchanger and re-entering the vessel was examined. Here, the same dependent variables were the focus; $U_i$, $h_i$, and $h_o$. For $U_i$, there was an overall non-linear downtrend but not much distinction in terms of the baffled vs unbaffled results. The baffled tank had a spike in its $U_i$ value at ~0.015 kg/s that quickly dropped once the impeller reached ~0.02 kg/s. The unbaffled tank showed a consistent decrease in $U_i$, with a sharp decrease at ~0.015 kg/s. Overall, at some points the $U_i$ value was higher for the baffled tank while at other points it was higher for the unbaffled tank. There wasn't a distinctive trend seen in this portion of the results.

Moving on to $h_i$, there was an almost linear-like increase at each recirculation flowrate. There was also a noticeable trend between the two tank configurations. For each flowrate measured, the baffled tank consistently had a higher $h_i$, while the unbaffled tank would be slightly lower. Overall, it seems that $h_i$ and the recirculation flowrate are directly correlated, but the baffled tank's $h_i$ seems to respond a little more to an increase in flow.

Looking at $h_o$, the results were similar to $h_i$ in that there was an overall increase in the value as the recirculation flowrate increased for both configurations. Although, the baffled tank involved a significant drop in the value of $h_o$ when the recirculation rate increased to 0.015 kg/s before eventually increasing. Unlike the results for $h_i$, there wasn't a trend regarding values being higher or lower for a baffled vs. an unbaffled tank. The results show numbers that are all over the place.

In terms of calculating all the variables in this section; $U_i$, $h_i$, and $h_o$ were all calculated the same as in the previous section. The recirculation flowrate was calculated by taking the reading from the respective trail and multiplying it by the maximum flowrate allowed by the vale. Next, the volumetric flowrate is turned into a mass flowrate by converting gallons to liters, multiplying by the density of water, and then finally converting minutes to seconds. All related graphs, tables, and sample calculations are mentioned in Appendix I & III, respectively.

**$Q_{13}$ vs. $U_i$, $h_i$, and $h_o$:**

Lastly, the effect of $Q_{13}$ on the respective dependent variables was measured. For $U_i$, the typical overall downtrend that has been common in this lab was seen again. For the baffled tank, it initially rises at ~4.90 kW but then falls as the heat entering the vessel reaches ~6.29 kW. The unbaffled tank shows an initial sharp drop in the value of $U_i$ at ~4.90 kW but only shows a very slight decrease once the finally heat flow is reached, ~6.29 kW. From the data, there is no correlation between a baffled or unbaffled tank and $Q_{13}$, the data is too random for both configurations

Looking at the effects of $Q_{13}$ on $h_i$, the results show that there is a direct correlation between $Q_{13}$ and $h_i$. The graph shows a positive linear-like relationship for the baffled and unbaffled tank configurations. It's also worth noting, that the $h_i$ values tend to be a tad bit higher for the baffled tank compared to the unbaffled at relatively the same heat flows.

Examining the results to determine how $Q_{13}$ correlates with $h_o$ led to the conclusion that there's no real relationship based on our data. Values were all over the place for both tank configurations and with there only being data for three different $Q_{13}$ for each vessel configuration, whatever overall trend is seen, can't be affixed to general interaction between $Q_{13}$ and $h_i$.

$U_i$, $h_i$, and $h_o$ were all calculated the same way as mentioned in the beginning section. Although, $Q_{13}$ was calculated using 'Method 2' which was mentioned in the lab manual [5]. This approach uses a heat balance around the vessel and assumes the system to be at steady state. This leads to the equation, $Q_{13} = Q_{hx} + Q_{coil} + Q_{surr}$. $Q_{surr}$, which is the heat loss from the recirculation piping and evaporation form the tank, is assumed to be negligible and given the value of zero. $Q_{hx} = m_{ves}C_p(t_{in} - t_{out})$ and $Q_{coil} = m_{coil}C_p(t_{out} - t_{in})$. Each mass flowrate is calculated by converting the volumetric flowrates given by the flowmeters for both the recirculation flow and the cooling coil flow. The heat capacity, $C_p$, is that of water. The temperature difference, ΔT, is the gradient seen across the coils and external heat exchanger, respectively.

Related graphs, tables, and sample calculations corroborating what has just been mentioned above, can be found in Appendix I and III.

## 5. Discussion

Concluding the experiment data overall is considered to be a mixed result in relation to theory from "*Transport Phenomena*". 2nd ed." In "*Transport Phenomena*". 2nd ed" the heat transfer should be decreasing when it is being influenced by cooling heat exchangers and stirring. The overall trend in h_i as both unbaffled and baffled trends would have an upward trend. But the issue comes in the Ui data analysis as human error and machine error can be accounted for issue in the analysis, as explained in the next paragraph. But overall, the Experimental data aligns with theoretical data of a downward trend in heat transfer as stirring or cooling occurs. In essence, theory is correct to the experiment when quantifying heat transfer in the experiment.

The results of the experiment revealed technical issues within the experimental machinery as complications have been constant throughout the experiment. An issue encountered while running the experiment was the external heat exchanger running efficiently after being cleaned. The cleaned heat exchanger caused issues with the system being too cool for the experiment temperature needed. This led to a long session of trouble shooting and trial and error in getting the approbative temperature needed in steady state. The complication with the heat exchanger led to the flow rate of both heat exchangers in the system staying fixed due to the slightest change in flow rate would cause the system to change in temperature at a large value. This would ruin consistency with data needed for analysis in the end. This complication happened in Trial 2 for Unbaffled and Baffled analysis, during trial 2 of the experiment the flow rate of the external heat exchanger was raised in order to keep a steady temperature but it cooled the system to quickly and steam was used to heat up the system raising the temperature and skewing the data upward for trial as seen in the graphs that contain Ui located in results. The baffled values for trial 2 are the only true discrepancy attributed to human error in steady state temperature management. This can be improved if a issue before the lab started was accounted for as the machine was recently cleaned up by the previous group. The external heat exchanger ran efficiently for the system, making changes requiring precision to establish a steady state environment for conducting the experiment. If that is applied, all data should follow the downward trend the graphs show by quantifying the data from the experiment.

The lack of general correlations for stirred tanks is due to the complicated geometry that must be accounted for from the stirred tanks. Normally, the general correlations for non-stirred and unbaffled tanks account for a simple design. In the case of the experiment, an alternative approach is needed for the creation of correlations for the stirred baffled tank. For the creation of new correlations for the stirred baffled tank, data is needed in the properties of the tank. Data needed for the correlations must cover the grounds of factors that complicate the system that is being measured. In an Impeller stirred baffled tank, the recorded values that are needed to set up the correlations are: Impeller Speeds, Temperature Changes, Properties, and Measured heat transfer coefficient for the experiment in different conditions. These values will set the basis for applying different conditions to the experiment that allow for the changes in correlations needed. Next, new correlations and parameters must be established to make correlations.

Parameters are needed to describe the system and apply constraints to dynamic data provided above. The needed parameters are the following: Nu (Nusselt Number), Pr (Prandtl Number), Re (Reynolds Number), Tank Size, Impeller Size, Baffle Size, Volume of the tank. With recorded data now used to help establish the bases for correlations.

A dimensional analysis method can be used to apply. Using generalized correlations for theory must be used to set up the bases for the new correlations. With the bases set now, the parameters and data can be used to establish the equation for the correlations. The analytical analysis for this experiment is to create a linear equation that uses parameters that can be associated with dynamic data that are recorded from the different conditions, for this lab the correlation will be a using Nu, Re Pr and the viscosity ratio.

This is written as:

Correlation Nu:

Correlation Pr:

Correlation Re

Correlation Viscosity Ratio:

List of values of internal compositions:

| Symbol | Description |
|---|---|
| | heat transfer coefficient (W/m²·K) |
| | impeller diameter (m) |
| | thermal conductivity of the fluid (W/m·K) |
| | fluid density (kg/m³) |
| | impeller rotational speed (revolutions/second) |
| | dynamic viscosity (kg/m·s) |
| | specific heat capacity (J/kg·K) |
| | kinematic viscosity (m²/s) |
| | thermal diffusivity (m²/s) |

| | |
|---|---|
| | Viscosity ratio exponent |
| | tank diameter (m) |
| | number of baffles |
| | baffle width (m) |

The new correlation equations can all be combined to create a overall Nu equation that can be used as a theoretical equation.

Now the equation must be linearly regressed for it to be applied to data analysis to lower the margin of error from theoretical and measured.

With the values for Correlated Nu values tend the measured Nu values standard error and R2 Analysis can be used in order to see how far off the experimental data is from the theory.
In conclusion, the Heat transfer lab results "*Unit Operations Laboratory: Heat Transfer to a Fluid in a Stirred Tank*". follow the theoretical trends that are stated "*Transport Phenomena*". 2nd ed.". The only issue being the machine error from the experiment limiting analysis from heat exchanger influence on the system and human error as setting a steady state for trial true leads to higher temperatures that corrupted the overall data trend. This leads to the following conclusion where more care must be taken when setting up the system in the right conditions to conduct the experiment in an orderly manner. Overall, the experiment is good in explaining the heat transfer phenomenon.

A remaining question that can be addressed is the effects of having a cooling through the heat exchangers provided. As the issues with machinery lead to limited analysis, from the flow rates in the heat exchangers.

A question that can help expand the experiment's phenomena is testing heat transfer with a reaction happening in the tank. This comes from having a CSTR reactor that is commonly used in the pharmaceutical industry. CSTR reactors are used for initial reactions that give off  alot of heat. Applying the same procedure with measuring heat transfer and the influence of heat exchanger and stirring can help to create an adaptable experiment.

## 6. References

[1] Fryer, P. J.; Robbins, P. T. Heat Transfer in Food Processing: Ensuring Product Quality and Safety. *Applied Thermal Engineering* **2005**, *25* (16), 2499–2510. https://doi.org/10.1016/j.applthermaleng.2004.11.021.

[2] Antognoli, M.; Qeysari, H.; Alberini, F.; Paglianti, A.; Singh, P.; Albano, A. Power and Fluid Stress in Disposable Square Stirred Tank to Streamline Pharmaceutical Process Development.

*International Journal of Pharmaceutics* **2025**, *685*, 126243. https://doi.org/10.1016/j.ijpharm.2025.126243.

[3] Bird, R. Byron, et al. *Transport Phenomena*. 2nd ed., John Wiley and Sons, Inc., 2002.

[4] Perry, Robert H., and Don W. Green, editors. *Perry's Chemical Engineering Handbook*. 8th ed., McGraw-Hill, 2008.

[5] Department of Chemical Engineering. 2025. *ChE 382: Unit Operations Laboratory: Heat Transfer to a Fluid in a Stirred Tank.* University of Illinois at Chicago.

## 7. Appendix I: Data Tabulation/Graphs

### Apparatus Parameters

| | | | |
|---|---|---|---|
| Impeller Rotation | Counter Clockwise | Max Coiling Cold Water Flowrate | 0.94 GPM |
| Vessel Outer Diameter | 0.35 m | Max Recirculation Flowrate | 1.1 GPM |
| Vertical Length of Tank | 0.49 m | | |
| Diameter of Impeller | 0.085 m | | |

### Trial #1 Data

Trial #1

Tank

| | |
|---|---|
| Impeller Speed | 300 RPM |

Unbaffled

| | |
|---|---|
| Impeller Speed | 268.4 RPM |
| Tank Temperature | 73.8 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 5 |
| T2 | 67.2 |
| T3 | 30.6 |
| T4 | 54 |
| T5 | 73 |
| T6 | 10.6 |
| T7 | 7.2 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.55 |
| Recirculation | 0.21 |
| Steam Condensate | 0.051 |

Baffled

| | |
|---|---|
| Impeller Speed | 340.9 RPM |
| Tank Temperature | 72 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 6.7 |
| T2 | 65.6 |
| T3 | 31.1 |
| T4 | 51 |
| T5 | 69 |
| T6 | 8.9 |
| T7 | 7.8 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.55 |
| Recirculation | 0.2 |
| Steam Condensate | 0.03 |

*Cooling coil water and recirculation flowrates are percentages of the maxiumum value

### Trial #2 Data

Trial #2

Tank

| | |
|---|---|
| Impeller Speed | 600 RPM |

Unbaffled

| | |
|---|---|
| Impeller Speed | 589.7 RPM |
| Tank Temperature | 70.8 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 5.6 |
| T2 | 65 |
| T3 | 35 |
| T4 | 44 |
| T5 | 69 |
| T6 | 10 |
| T7 | 6.1 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.564 |
| Recirculation | 0.077 |
| Steam Condensate | 0.018 |

Baffled

| | |
|---|---|
| Impeller Speed | 614.3 RPM |
| Tank Temperature | 69.1 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 5.6 |
| T2 | 62.2 |
| T3 | 33.3 |
| T4 | 45 |
| T5 | 67 |
| T6 | 12.8 |
| T7 | 6.7 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.55 |
| Recirculation | 0.14 |
| Steam Condensate | 0.06 |

**Trial #3 Data**

Trial #3

Tank

| | |
|---|---|
| Impeller Speed | 900 RPM |

Unbaffled

| | |
|---|---|
| Impeller Speed | 883.3 RPM |
| Tank Temperature | 65.8 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 7.2 |
| T2 | 60 |
| T3 | 34,4 |
| T4 | 34 |
| T5 | 45 |
| T6 | 11.1 |
| T7 | 9.4 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.84 |
| Recirculation | 0.088 |
| Steam Condensate | 0.017 |

Baffled

| | |
|---|---|
| Impeller Speed | 614.3 RPM |
| Tank Temperature | 69.1 ° C |

| Gauge Number | Temperature (°C) |
|---|---|
| T1 | 6.1 |
| T2 | 61.1 |
| T3 | 35.6 |
| T4 | 49 |
| T5 | 67 |
| T6 | 10 |
| T7 | 7.8 |

| Flow Type | Volume flow rate (GPM) |
|---|---|
| Cooling Coil Water | 0.55 |
| Recirculation | 0.21 |
| Steam Condensate | 0.02 |

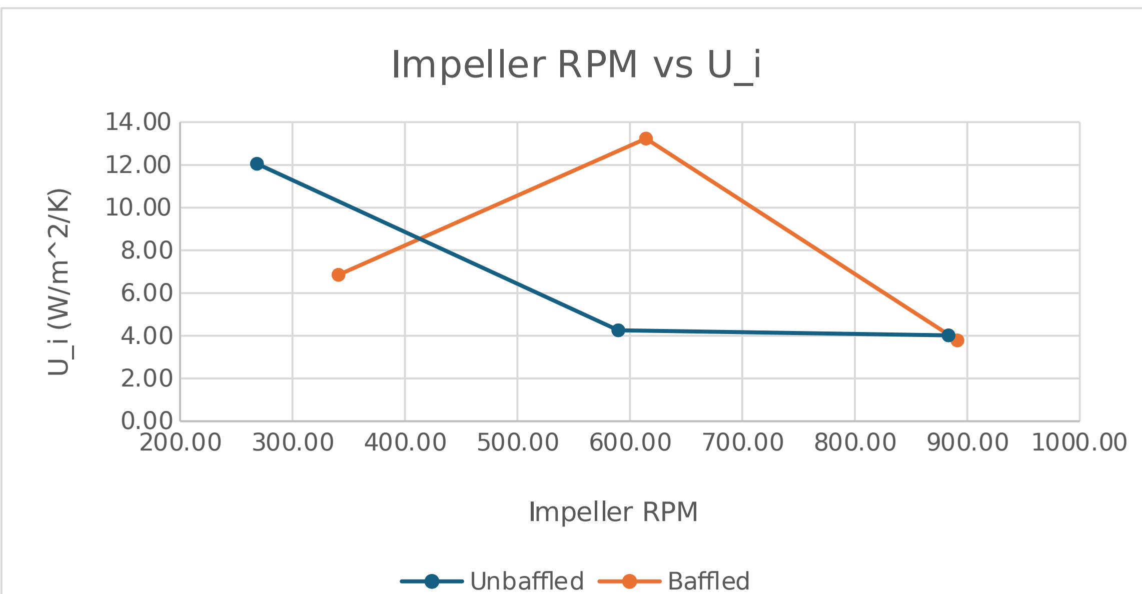
**Trial Calculations**

Unbaffled

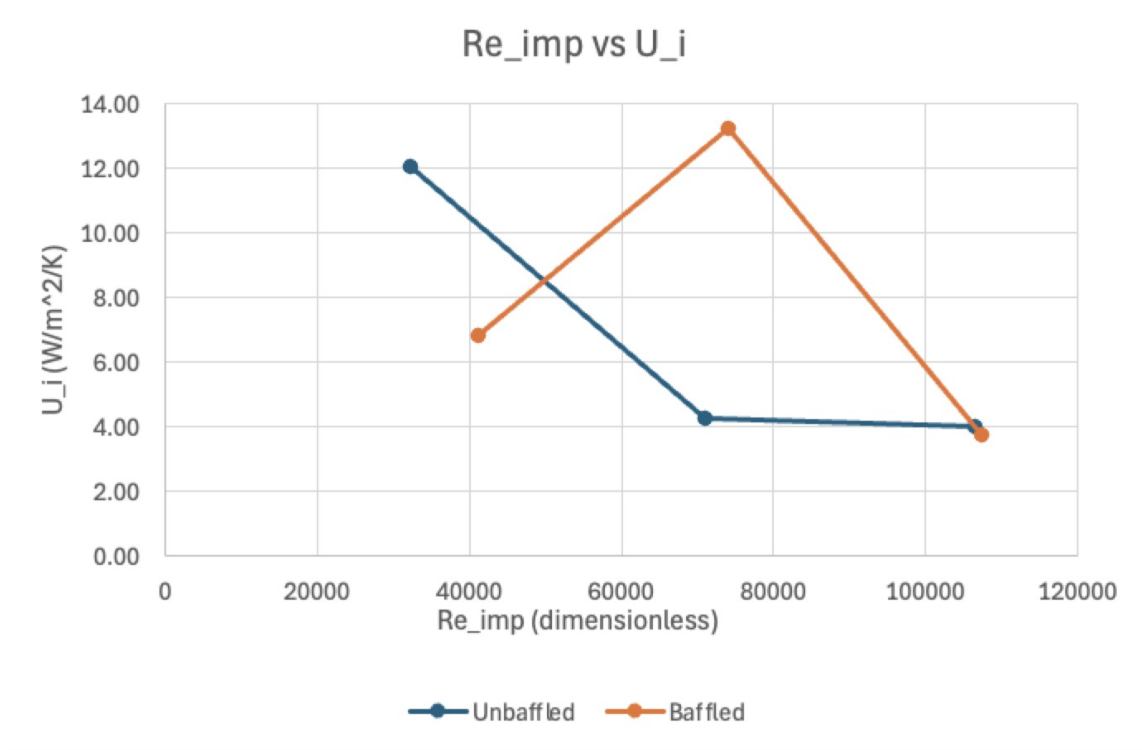
| Trial | 1 | 2 | 3 |
|---|---|---|---|
| m_ves (kg/s) | 0.0136 | 0.017 | 0.021 |
| m_coil (kg/s) | 0.0351 | 0.0357 | 0.053 |
| ΔT_ves (K) | 19 | 25 | 11 |
| ΔT_coil (K) | 25.6 | 29.4 | 27.2 |
| Q_hx (kW) | 1.08 | 1.78 | 0.97 |
| Q_coil (kW) | 3.76 | 4.39 | 6.03 |
| Q_13 (kW) | 4.84 | 6.17 | 7 |
| m_sc (kg/s) | 0.0032 | 0.0011 | 0.0011 |
| T_steam (K) | 1086.2 | 3031.7 | 3566.1 |
| U_i (W/m^2/K) | 12.05 | 4.25 | 4.02 |
| h_o (W/m^2/K) | 8295.6 | 11738.6 | 11964.4 |
| *Γ* (kg/m/s) | 0.00294 | 0.00104 | 0.00098 |
| h_i (W/m^2/K) | 12.07 | 4.25 | 4.02 |
| RPS (rev/s) | 4.47 | 9.83 | 14.72 |
| Re_imp | 32296 | 71022 | 106352 |
| Pr | 0.007 | 0.007 | 0.007 |
| Nu_T | 69.8 | 118 | 154.4 |
| h_i (W/m^2/K) | 119.6 | 202.2 | 264.7 |

Baffled

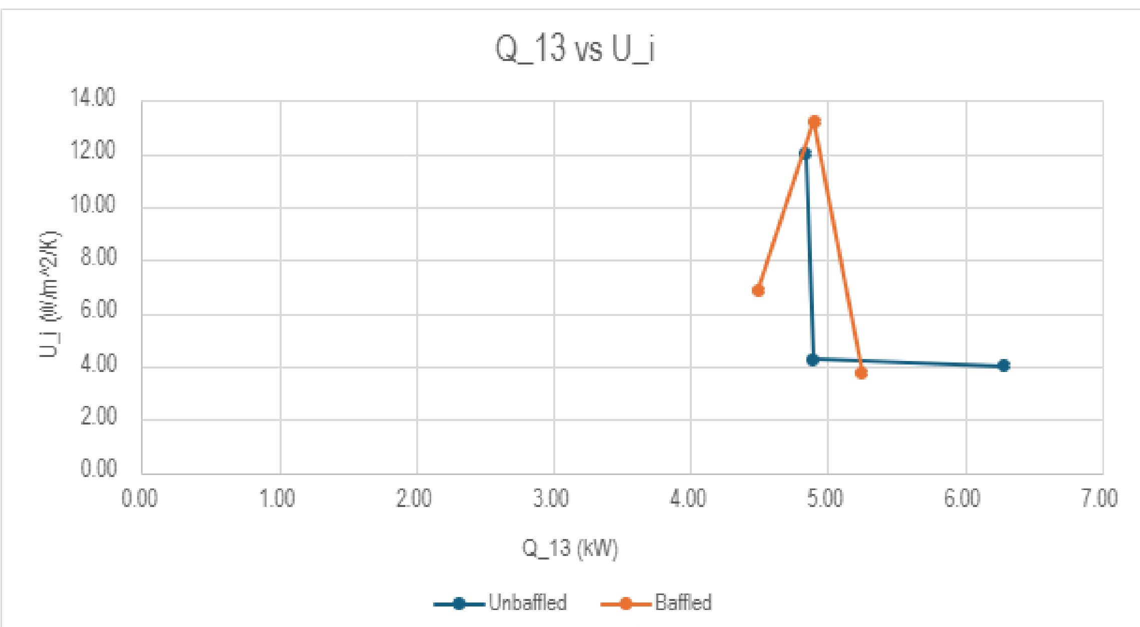
| Trial | 1 | 2 | 3 |
|---|---|---|---|
| m_ves (kg/s) | 0.0129 | 0.0156 | 0.0204 |
| m_coil (kg/s) | 0.0345 | 0.0351 | 0.0345 |
| ΔT_ves (K) | 18 | 22 | 18 |
| ΔT_coil (K) | 24.4 | 27.7 | 29.5 |
| Q_hx (kW) | 0.97 | 1.44 | 1.54 |
| Q_coil (kW) | 3.53 | 4.07 | 4.26 |
| Q_13 (kW) | 4.5 | 5.51 | 5.8 |
| m_sc (kg/s) | 0.0018 | 0.0035 | 0.001 |
| T_steam (K) | 1556.5 | 1107.8 | 3181 |
| U_i (W/m^2/K) | 6.85 | 13.23 | 3.78 |
| h_o (W/m^2/K) | 10013.2 | 8041 | 12208.6 |
| *Γ* (kg/m/s) | 0.00167 | 0.00323 | 0.00092 |
| h_i (W/m^2/K) | 6.86 | 13.25 | 3.78 |
| RPS (rev/s) | 5.68 | 10.24 | 14.85 |
| Re_imp | 41038 | 73984 | 107291 |
| Pr | 0.007 | 0.007 | 0.007 |
| Nu_T | 81.8 | 121.2 | 155.3 |
| h_i (W/m^2/K) | 140.3 | 207.8 | 266.2 |



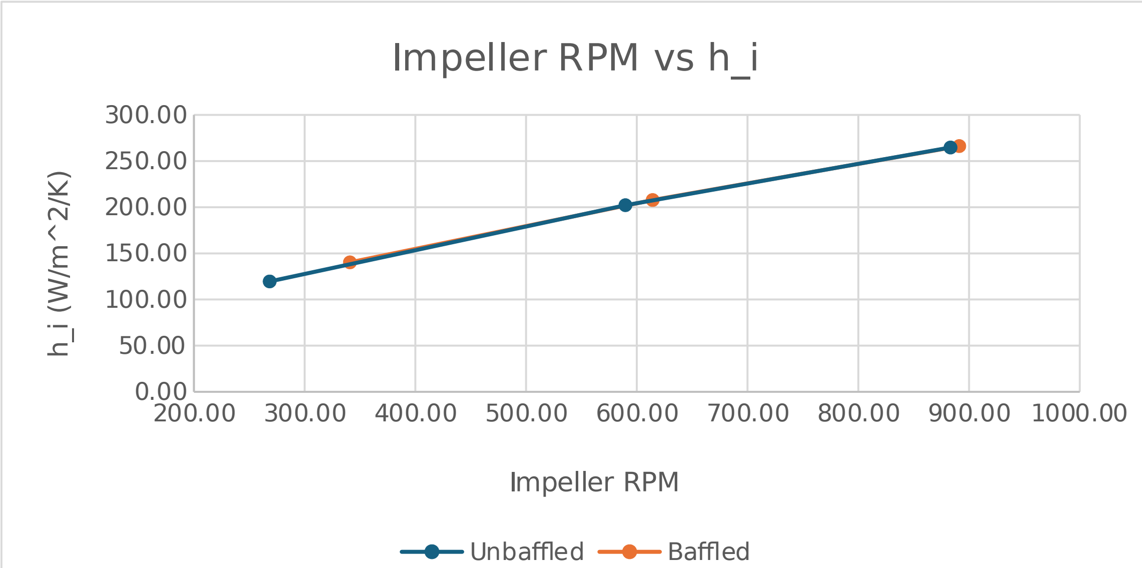

Impeller RPM vs


Re_imp vs U_i
U_i (W/m^2/K)
Re_imp (dimensionless)
14.00
12.00
10.00
8.00
6.00
4.00
2.00
0.00
0
20000
40000
60000
80000
100000
120000
Unbaffled
Baffled

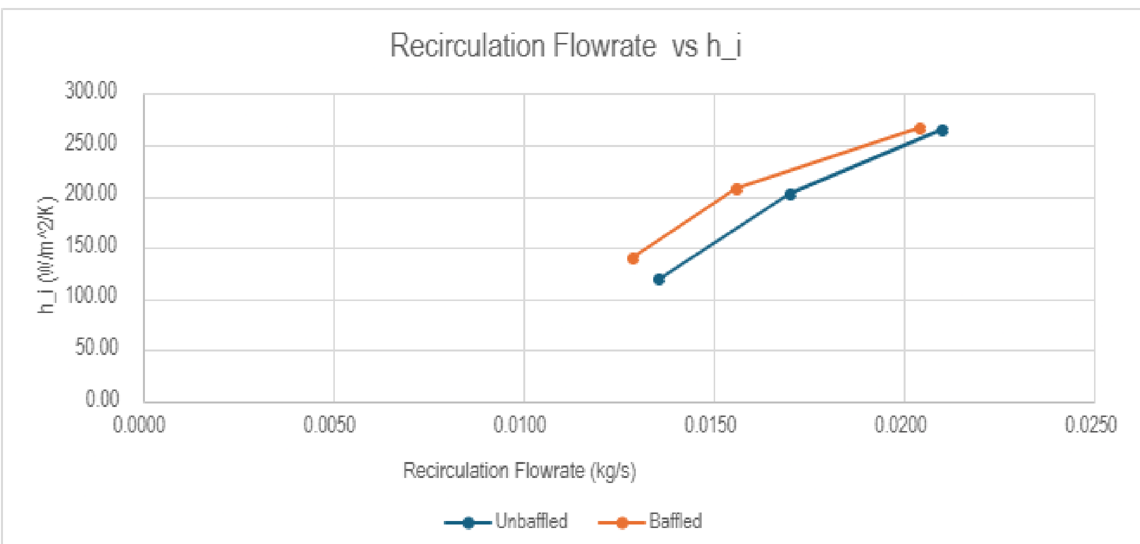
vs


Q_13 vs U_i
U_i (W/m^2/K)
Q_13 (kW)
14.00
12.00
10.00
8.00
6.00
4.00
2.00
0.00
0.00
1.00
2.00
3.00
4.00
5.00
6.00
7.00
Unbaffled
Baffled

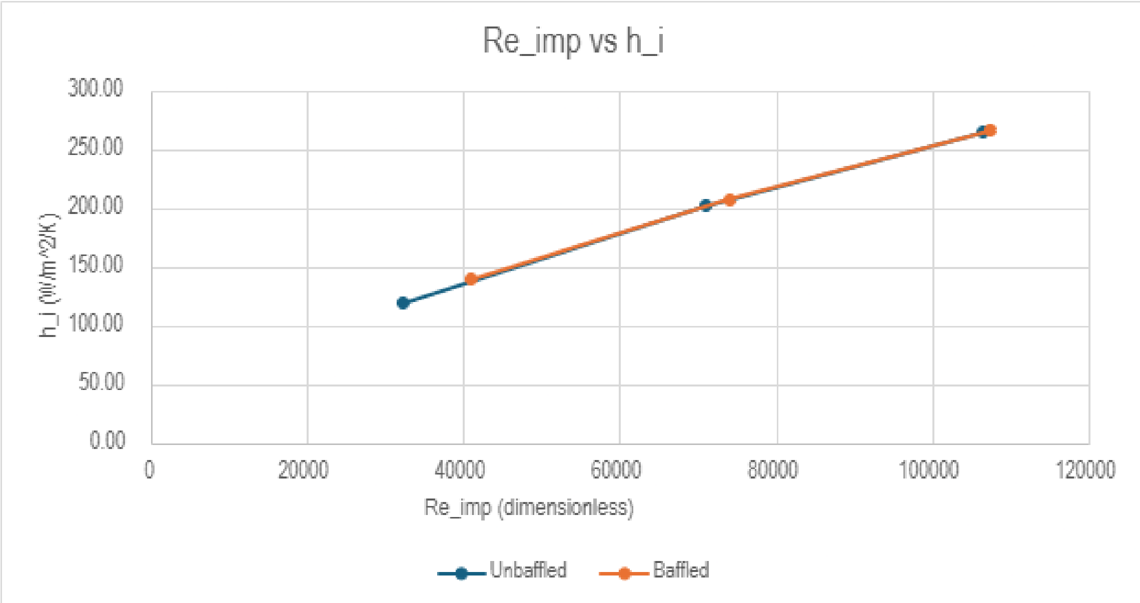
$Q_{13}$ vs $U_i$



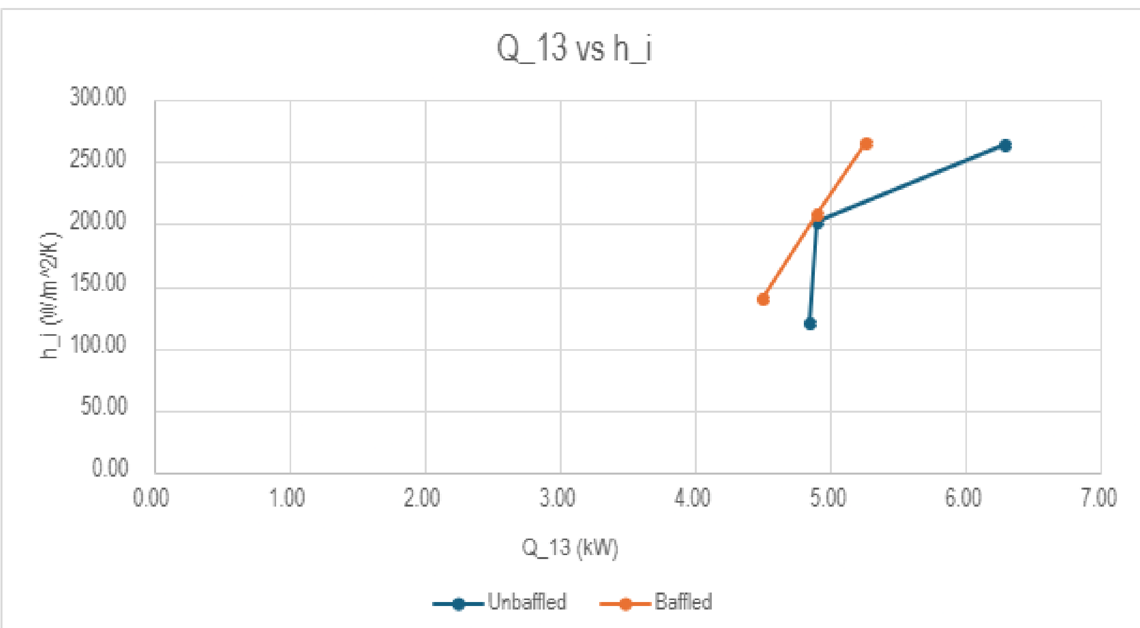

Impeller RPM vs



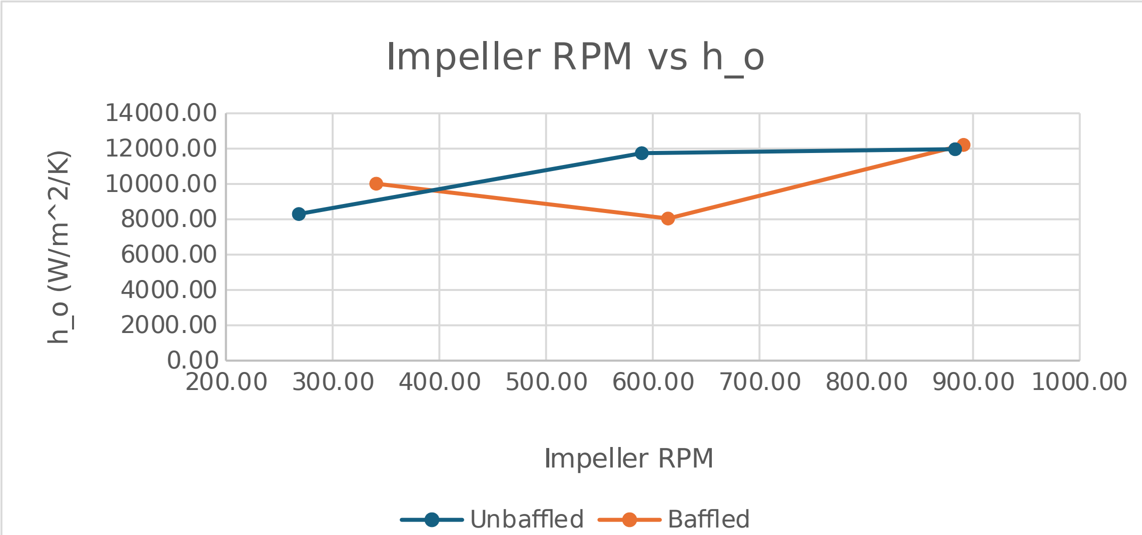

Recirculation Flowrate vs



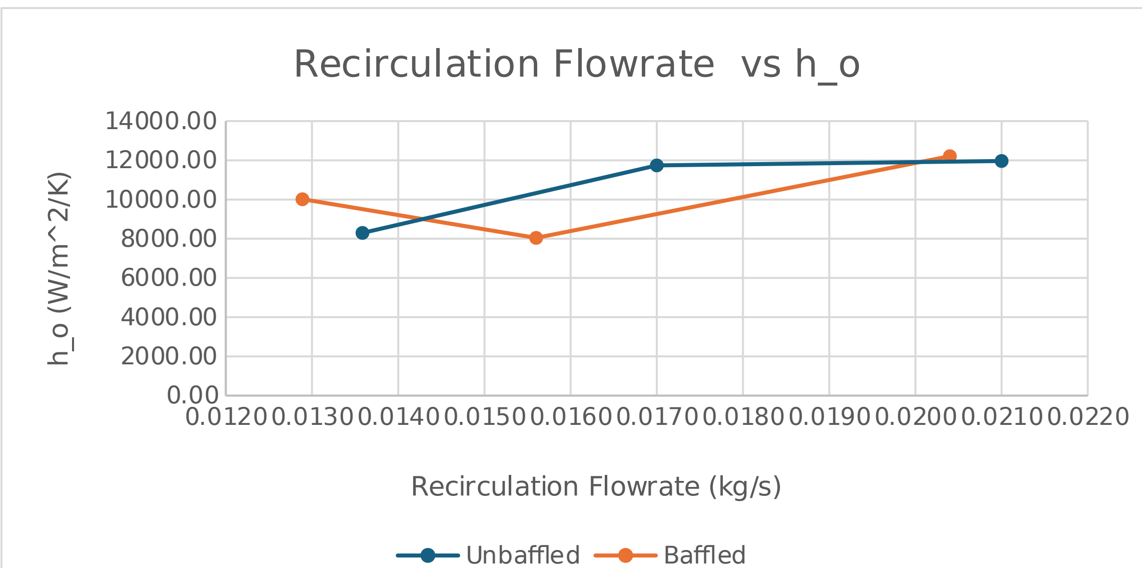

$Re_{imp}$ vs $h_i$



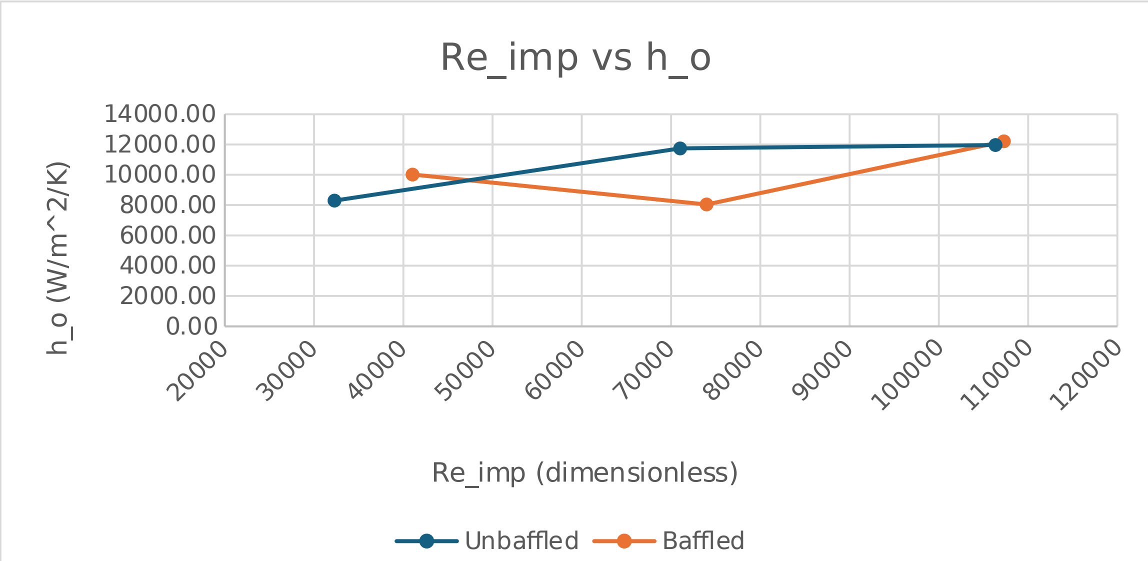

vs $h_i$



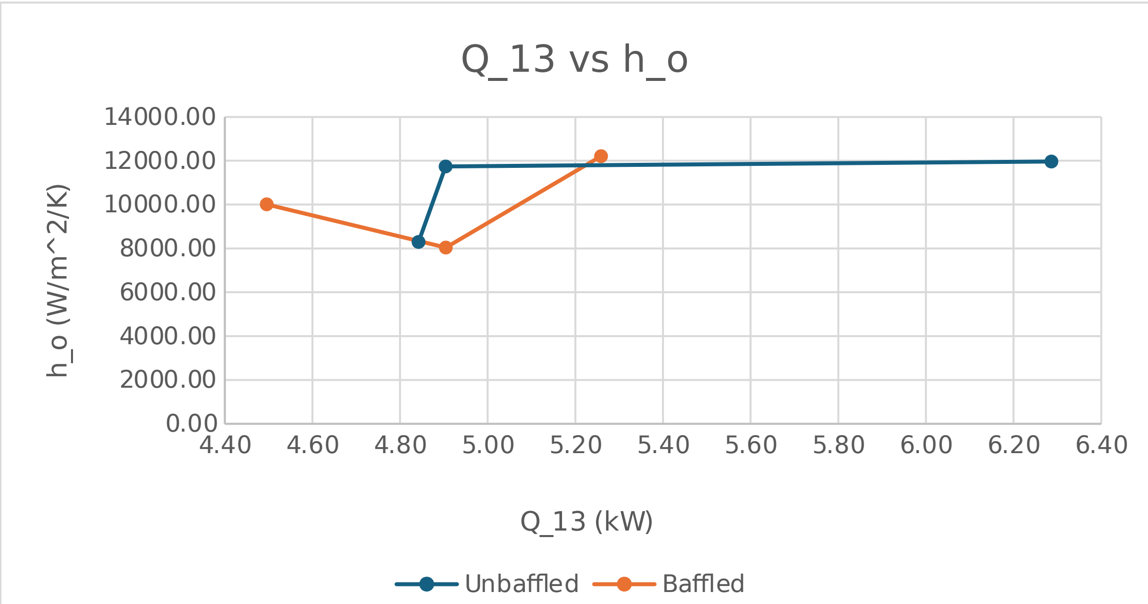

Impeller RPM vs




Recirculation Flowrate vs




vs




vs

## 8. Appendix II: Error Analysis

The results for the heat transfer coefficients involved several sources of uncertainty that impacted the accuracy of the experimental and final calculations. For the quantitative error analysis, the most significant source and variance happened during the measurement of the impeller, speed, the tachometer readings were seen to vary approximately 25% during steady state operations. Due to the theoretical inner wall heat transfer coefficient is derived from the impeller Reynolds number ($Re_{imp}$) depends on the rotational speed. This 25% fluctuation introduces a large margin of error in the predicted Nusselt number. Going on, the rotameters used to measure the recirculation flow rates and cooling coil show a hold-up effect, where the float failed to return to its original position which results in a consistent 0.5 LPM offset. This inconsistency affects the mass flow rate values used in the energy balance equations, leading to inaccuracies in the calculated total heat transfer rate (*Heat Transfer Lab Manual Page 4,5,7-8*).

For the qualitative error analysis, the reliability of the steam energy (Method 1) had reduced as the condensate water didn't always leave the system at a steady-state rate. There was intermittent flow and signs of clogging in the steam trap meaning that the collection of condensation over time didn't accurately show the true rate of steam condensation. Also, the formation of a vortex at a high impeller speed during the unbaffled trials possibly altered the effective heat transfer area, causing deviations from theoretical correlations. Finally, assuming negligible heat loss from the uninsulated piping and the lid could cause the energy balance of the fluid to underplay the heat transferred from the steam jacket (*Heat Transfer Lab Manual Page 4,6,9*).

## 9. Appendix III: Sample Calculations

**Trial #1** (Unbaffled)

**Impeller Speed**: 268.4 RPM

$V_{ccw}$ = 0.55 GPM (cooling-coil water), $V_{rf}$ = 0.21 GPM (recirculation flow), and
$V_{sc}$ = 0.051 GPM (steam condensate)

$T_1$ = 5 ℃, $T_2$ = 67.2 ℃, $T_3$ = 30.6 ℃, $T_4$ = 54 ℃, $T_5$ = 73 ℃, $T_6$ = 10.6 ℃, and
$T_7$ = 7.2 ℃

**Calculation of :** The total heat transferred into the vessel is composed of three contributions:

Where:

- *Assumptions*: no scale on either side of the tank, high k value for vessel, and small wall thickness
- : Heat loss by the recirculated fluid passing through the heat exchanger
- : Heat gained by the fluid in the cooling coil

- : Heat loss to the surroundings (assumed to be zero)

**Heat Exchanger Contribution ():**

Where:

- (mass flow rate of recirculated fluid)
- (specific heat of water)
- : Temperature of fluid entering the heat exchanger in K
- : Temperature of fluid leaving the heat exchanger in K

**Cooling Coil Contribution ():**

Where:

- (mass flow rate of cooling coil water)
-
- : increase in temperature of fluid in cooling coils

***Recirculated Fluid (:***

-
- Mass flow rate of recirculated fluid:

***Cooling Coil Water (:***

-
- Mass flow rate of cooling coil water:

**Heat Exchanger Contribution ():**

**Cooling Coil Contribution ():**

**Total Heat Transfer ():**

**Calculating Overall Heat Transfer Coefficient ():**

Using the relation:

**Surface Area of Heat Exchange ():**

- is the outer diameter of the vessel
- is the vertical length of the tank

**Alternate Expression for  Using Steam Condensate:**

- is the temperature of the steam and is the temperature of the water in the vessel; it corresponds with $T_2$ in our data.
- was found using Perry's Chemical Engineering Handbook and searching for steam's heat capacity at 100 ℃ as a good estimate for the lab.

**Mass Flow Rate of Steam Condensate ():**

**Calculating Steam Temperature ():**

**Calculating Overall Heat Transfer Coefficient ():**

**Calculating External Heat Transfer Coefficient ():**

Where:

- (vertical length of tank)
- (thermal conductivity of water)
- (density of water)
- (viscosity of water)
- (gravitational acceleration)
-
- (outer diameter of tank)
- (Steam condensate flowrate)

*Note: Approximate values sourced from NIST & Perry's Chemical Engineering Handbook when needed*

**Film Reynolds Number ():**

*Note: This value is not greater than 2100*

**External Heat Transfer Coefficient ():**

Substituting values:

**Internal Heat Transfer Coefficient (): [calculating using both equations (4) and (9) from lab handout]**

Equation (4):

Given:

-

- 

Substitute:

**Equation (9):**

**Impeller Reynolds Number ():**

*Note: 268.4 RPM = 4.447 RPS*

*Where:*

- is the viscosity of water ()
- is the impeller diameter (meters)
- is the revolutions of the impeller (revolutions per second)

**Calculating Prandtl Number ()**:

**Calculating Nusselt Number ()**:

**Calculating Internal Heat Transfer Coefficient ():**

Using the Nusselt relation:

## 10. Appendix IV: Individual Team Contributions

| | **TIME (HOURS)** | **DESCRIPTION** |
|---|---|---|
| **Operator (Both Lab Days)** | 5 hours | |

| | | |
|---|---|---|
| **Pre-Lab Editing** | 30 minutes | Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu |
| **Experimental Objective (Prelab)** | 15 minutes | Joshua Laforest |
| **Apparatus (prelab)** | | Javier Silva |
| **Materials & Supplies (prelab)** | 45 minutes | Ify Bidokwu |
| **Procedure (prelab)** | 45 minutes | Cesearo Dilosa |
| **Project Safety Evaluation (prelab)** | 65 minutes | Ify Bidokwu |
| **Final Lab Editing** | 45 minutes | Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu |
| **Executive Summary (final lab)** | 30 minutes | Joshua Laforest |
| **Introduction (Final lab)** | 50 minutes | Joshua Laforest |
| **Theory (Final Lab)** | 60 minutes | Ify Bidokwu |
| **Results (final lab)** | 75 minutes | Cesearo Dilosa |
| **Discussion (final lab)** | 3 hours | Javier Silva |
| **References (final and/or prelab)** | 15 minutes | Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu |
| **Data Tabulation/Graphs (final)** | 60 minutes | Cesearo Dilosa, Joshua Laforest |
| **Error Analysis (final lab)** | 45 minutes | Ify Bidokwu |
| **Sample Calculations (final Lab)** | 60 minutes | Cesearo Dilosa |
| **Power Point Presentation** | 45 minutes | Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu |
| **Total** | 20 hours 25 minutes | |